# Deployment documentation and user's manual

PosePro Application

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# 1 Deployment documentation and installation instruction

This section covers the installation and building process of the PosePro application using the remote GitHub repository. All necessary requirements for installing the application are described below.

**Operating system:** Windows 10/11 (or newer) or a Linux-based operating system
**Development environment:** Python, recommended version: 3.12.0

1. First, install and clone the remote repository from GitHub using the following command in the terminal:

   ```
   git clone git clone https://github.com/Michael-Pytel/PosePro.git
   ```

2. Next, navigate to the project directory, create a virtual environment and run it using the following instructions:

   ```
   cd PosePro
   python -m venv venv
   .\venv\Scripts\activate
   ```

3. Install all necessary dependencies using the command:

   ```
   pip install -r requirements.txt
   ```

4. Check if you have the ffmpeg command line tool installed using this instruction:

   ```
   ffmpeg -version
   ```

   If not, use one of the following methods to install:
   **Method 1 (Windows winget installation):**

   ```
   winget install Gyan.FFmpeg
   ```

   If you don't have the winget tool, install it. For example:

   (a) Go on: `https://github.com/microsoft/winget-cli/releases/lates`
   (b) Find this file and download it: `Microsoft.DesktopAppInstaller_8wekyb3d8bbwe.msixbundle`
   (c) Install it and restart your Command Line tool.

   **Method 2 (Chocolatey Windows installation):**

   ```
   choco install ffmpeg -y
   ```

   In case you don't have the chocoloatey tool, open powershell as an administrator and type those commands:

```
Set-ExecutionPolicy Bypass -Scope Process -Force

[System.Net.ServicePointManager]::SecurityProtocol =
    [System.Net.ServicePointManager]::SecurityProtocol -bor 3072

$url = 'https://community.chocolatey.org/install.ps1'
iex ((New-Object System.Net.WebClient).DownloadString($url))
```

**Method 3 (Linux Ubuntu/Debian installation):** Run the following commands:

```
sudo apt update
sudo apt install ffmpeg -y
ffmpeg -version
```

**Method 4 (Linux snap installation):** Execute those instructions:

```
sudo snap install ffmpeg
ffmpeg -version
```

5. Perform the necessary migrations executing those commands:

```
python manage.py makemigrations
python manage.py migrate
```

6. Run the application using this instruction:

```
python manage.py runserver
```

After successfully executing the command, the local server address should be displayed in the console. This is most often: `http://127.0.0.1:8000`. If installation went as planned, to restart or run the application once again, simply use the command shown in step 6.

## 2 User's Manual

After launching the application, the main view appears, which briefly describes the application and its capabilities. This is shown in Figure 1.

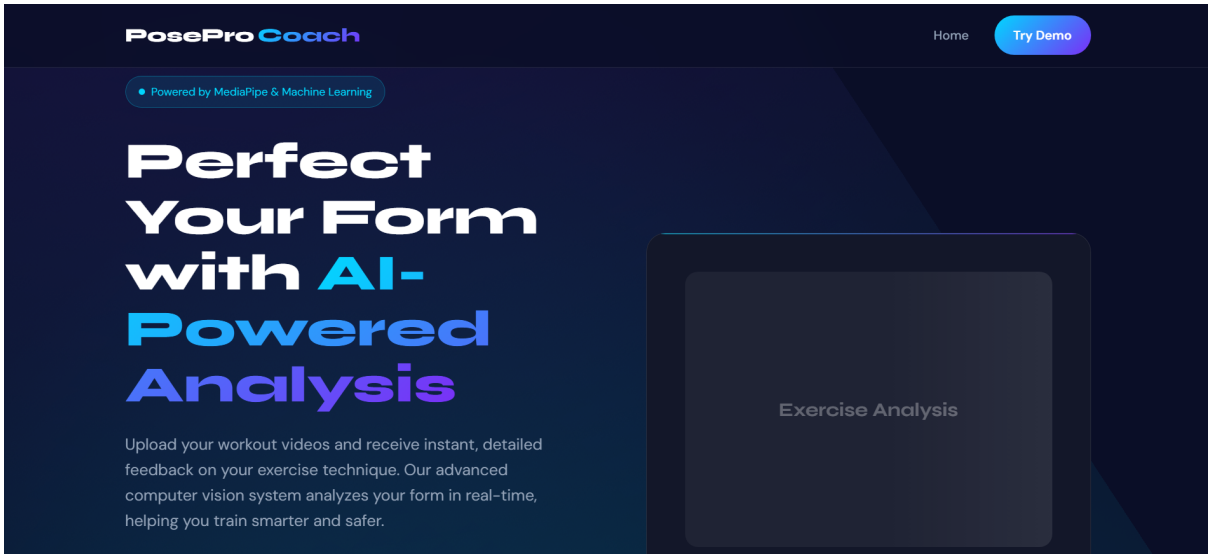


Figure 1: Main view of the application

In order to use the main functionality of the application, the user can click the "Try demo" button, which is located in the upper right corner and also in the lower part of the main page. After clicking the button, the user is presented with the view shown in Figure 2.

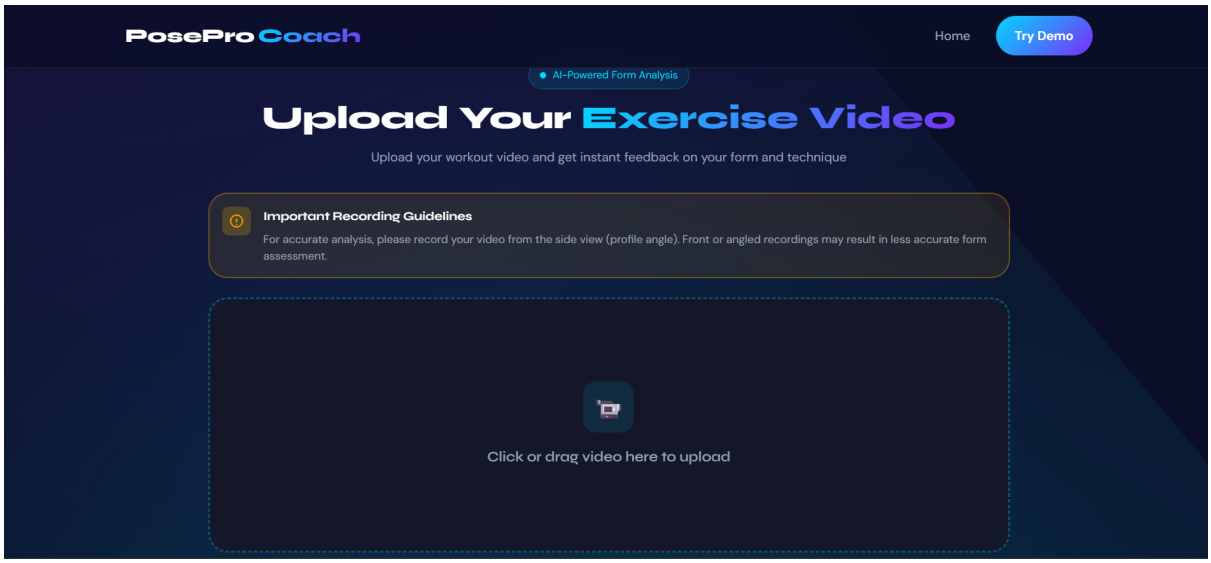


Figure 2: Main system functionality view

Next, upload the desired video by clicking the "Click or drag video here to upload" button or simply dropping the video. Please note that the video must be in one of the following formats: MP4, MOV, or AVI and cannot exceed 150 MB in size. After uploading the video, the view should look similar to the one shown in Figure 3.

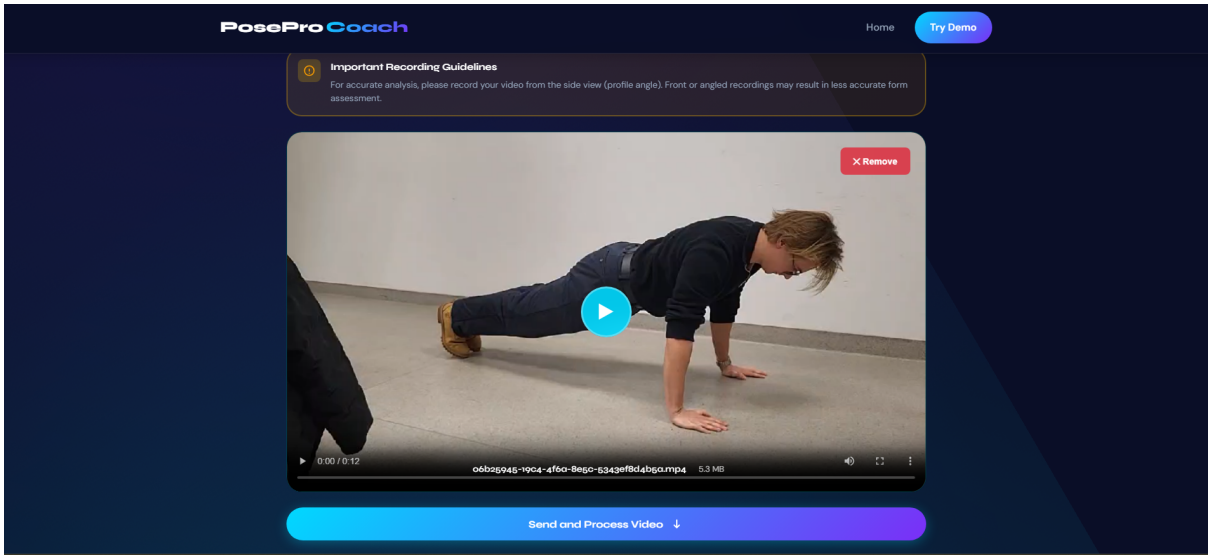


Figure 3: Uploaded video view

Next, click the "Send and process video" button, which should display the view shown in Figure 4.

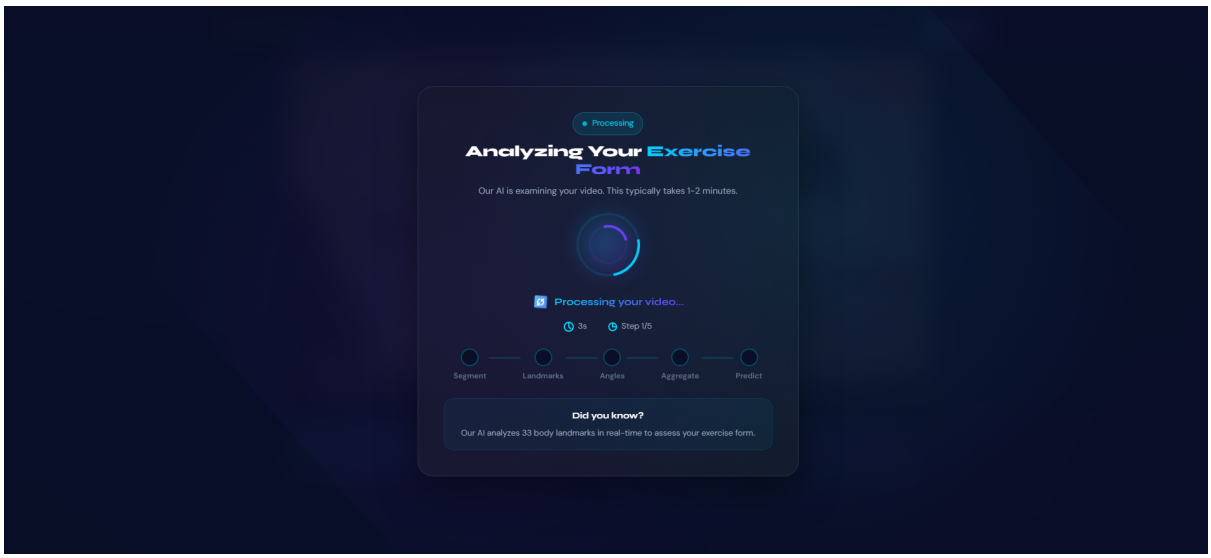


Figure 4: Video processing view

After the video has been successfully processed, the view showing the analysis results should appear, as shown in Figure 5.

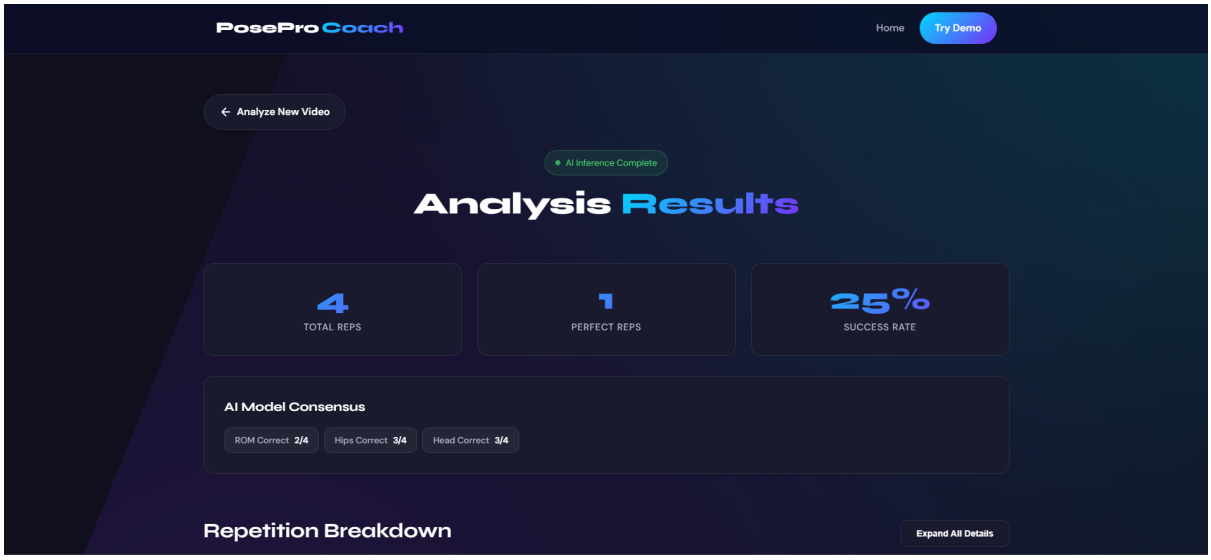


Figure 5: Biomechanical analysis view

You can verify the accuracy of each repetition and check the biomechanical parameters, including the values of individual angles and other interesting characteristics. To do that, you need to click "Show details" button of the repetition you are interested in biomechanical analysis. Sample visualizations are shown in Figures 6 and 7.

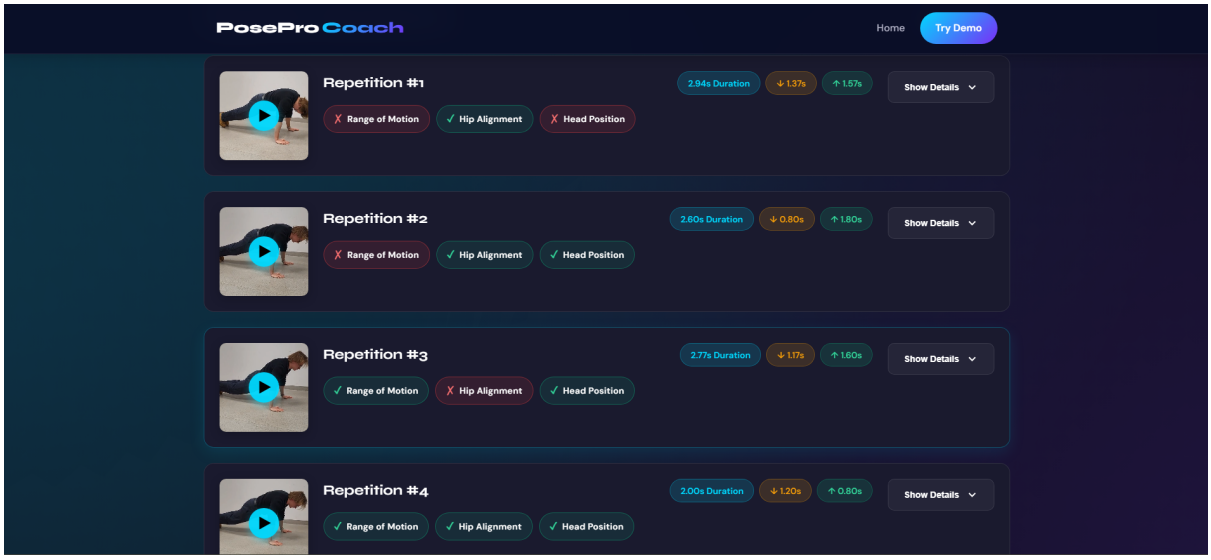


Figure 6: Repetition analysis visualization

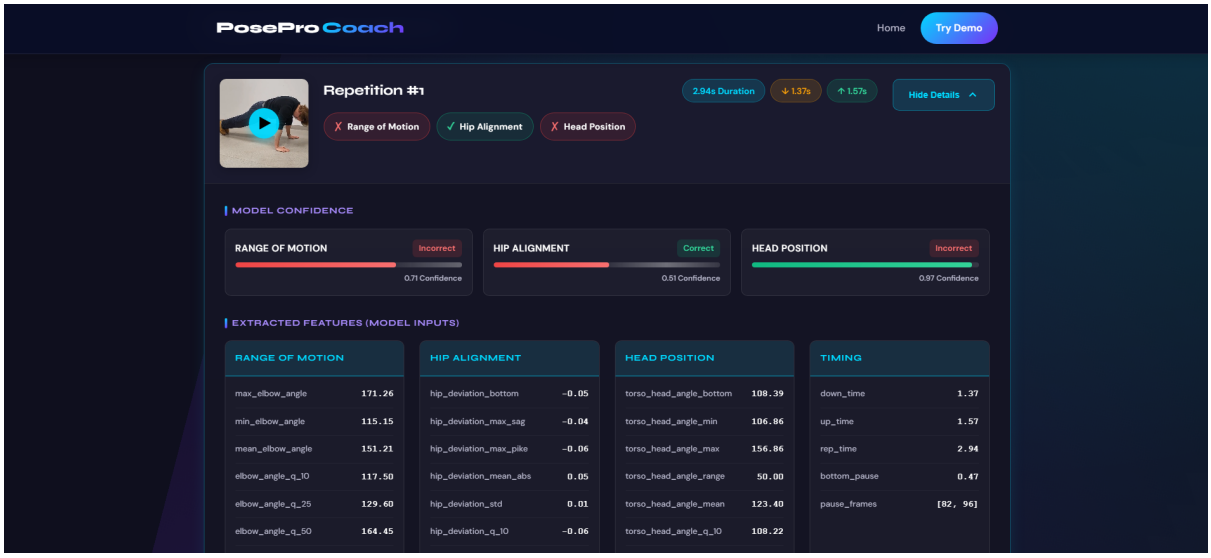


Figure 7: Sample biomechanical characteristics visualization